TITLE: Analyzing Index Performance Impact

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INTRODUCTION

This lab focuses on understanding how database indexes affect query performance. The main goal is to see how indexes help speed up data retrieval by reducing the number of rows that need to be scanned. By comparing queries with and without indexes, this activity shows why indexes are important in real-world database systems and how they improve efficiency.

PROCEDURE

First, I created a database named *index_performance_lab* and a table called *users* with columns for id, username, email, and created_at. After that, I inserted 100,000 rows into the table using a stored procedure.

Next, I ran a query to search for a specific username without using any index and recorded its execution time. I then used the EXPLAIN command to analyze how the query was executed.

After that, I created an index on the username column and ran the same query again to compare the performance. I also checked the execution plan again using EXPLAIN.

Lastly, I tested other queries using LIKE and conditions on the id column to observe how indexes behave in different situations.

RESULTS

The query without an index took more time because the database performed a full table scan. After creating the index, the same query became significantly faster.

The EXPLAIN results showed that before indexing, the query type was “ALL,” meaning it scanned the entire table. After indexing, the type changed to “ref,” and the index *idx_username* was used.

ANALYSIS

The index improved query performance by allowing the database to quickly locate the needed row instead of scanning all records. This is especially useful for large datasets like the one used in this lab.

However, indexes also have limitations. They take up extra storage and can slow down insert and update operations because the index also needs to be updated.

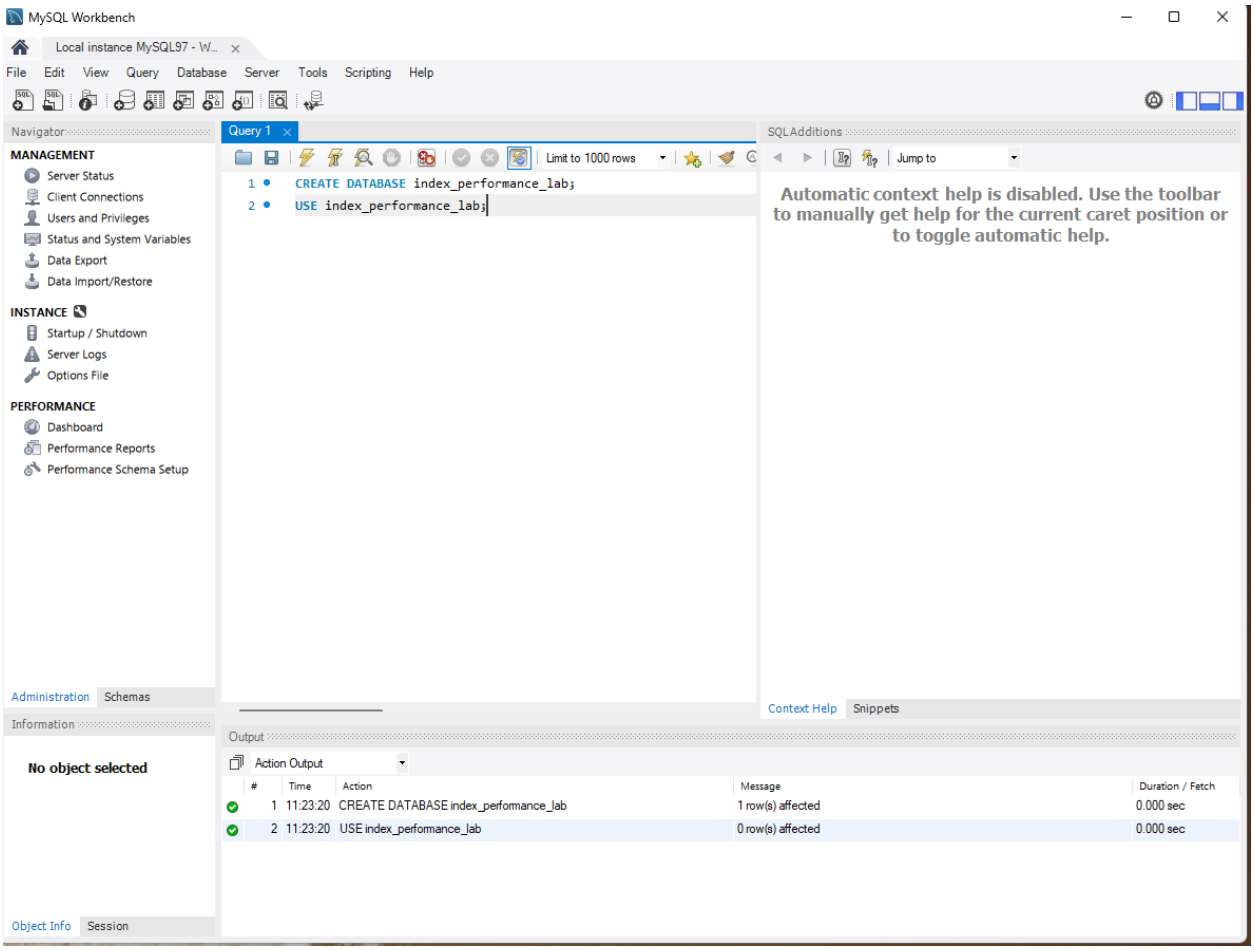
For the LIKE queries, I noticed that patterns like ‘user5%’ used the index, while patterns like ‘%user5’ did not. This is because indexes work best when the search starts from the beginning of the string.

Overall, indexes are very useful, but they should be used carefully depending on the type of queries.

CONCLUSION

In this lab, I learned how indexes can greatly improve query performance by reducing the amount of data scanned. I also understood how to analyze execution plans using EXPLAIN and identify whether an index is being used.

Additionally, I learned that indexes are not always beneficial in every case, especially with certain query patterns or frequent data updates. This lab helped me understand when and why to use indexes in a database.



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Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
CREATE TABLE users
(
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(255) NOT NULL,
    email VARCHAR(255) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

SQLAdditions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:23:20	CREATE DATABASE index_performance_lab	1 row(s) affected	0.000 sec
2	11:23:20	USE index_performance_lab	0 row(s) affected	0.000 sec
3	11:25:37	CREATE TABLE users (id INT AUTO_INCREMENT PRIMARY KEY, us...	0 row(s) affected	0.015 sec

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Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
DELIMITER //
CREATE PROCEDURE insert_users(IN num_rows INT)
BEGIN
DECLARE i INT DEFAULT 1;
WHILE i <= num_rows DO
INSERT INTO users (username, email)
VALUES (concat('user', i), concat('users', i, '@example.com'));
SET i = i + 1;
END WHILE;
END //
DELIMITER ;

CALL insert_users(100000);
```

SQLAdditions

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4	11:29:54	CREATE PROCEDURE insert_users(IN num_rows INT) BEGIN DECLARE i ...	0 row(s) affected	0.016 sec
5	11:29:54	CALL insert_users(100000)	1 row(s) affected	106.266 sec

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Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1  SELECT * FROM users WHERE username = 'user50000';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

id	username	email	created_at
50000	user50000	users50000@example.com	2026-04-25 11:30:46
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

users 2

Apply Revert

SQLAdditions

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Context Help Snippets

Output

Action Output

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5	11:29:54	CALL insert_users(100000)	1 row(s) affected	106.266 sec
6	11:37:24	SELECT * FROM users WHERE username = 'user5000' LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
7	11:37:53	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.032 sec / 0.000 sec

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File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1 •  SELECT * FROM users WHERE username = 'user5000';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

id	username	email	created_at
5000	user5000	users5000@example.com	2026-04-25 11:30:00
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

users 1

Apply Revert

SQLAdditions

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Context Help Snippets

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6	11:37:24	SELECT * FROM users WHERE username = 'user5000' LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

MySQL Workbench

Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1  EXPLAIN SELECT * FROM users WHERE username = 'user50000';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EXPLAIN
-> Filter: (users.username = 'user50000') (cost...

Result Grid | Form Editor | Field Types | Query Stats

Result 3 | Read Only

SQL Additions

Jump to

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Context Help Snippets

Output

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5	11:29:54	CALL insert_users(100000)	1 row(s) affected	106.266 sec
6	11:37:24	SELECT * FROM users WHERE username = 'user5000' LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
7	11:37:53	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.032 sec / 0.000 sec
8	11:38:04	EXPLAIN SELECT * FROM users WHERE username = 'user50000'	1 row(s) returned	0.015 sec / 0.000 sec

MySQL Workbench

Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1  CREATE INDEX idx_username ON users (username);
```

SQL Additions

Jump to

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Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	11:29:54	CREATE PROCEDURE insert_users(IN num_rows INT) BEGIN DECLAR...	0 row(s) affected	0.016 sec
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7	11:37:53	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.032 sec / 0.000 sec
8	11:38:04	EXPLAIN SELECT * FROM users WHERE username = 'user50000'	1 row(s) returned	0.015 sec / 0.000 sec
9	11:38:25	CREATE INDEX idx_username ON users (username)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.156 sec

MySQL Workbench

Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1   SELECT * FROM users WHERE username = 'user50000';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

id	username	email	created_at
50000	user50000	users50000@example.com	2026-04-25 11:30:46
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

users 4 | Apply | Revert

SQLAdditions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	11:29:54	CALL insert_users(100000)	1 row(s) affected	106.266 sec
6	11:37:24	SELECT * FROM users WHERE username = 'user5000' LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
7	11:37:53	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.032 sec / 0.000 sec
8	11:38:04	EXPLAIN SELECT * FROM users WHERE username = 'user50000'	1 row(s) returned	0.015 sec / 0.000 sec
9	11:38:25	CREATE INDEX idx_username ON users (username)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.156 sec
10	11:38:42	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1 •  EXPLAIN SELECT * FROM users WHERE username = 'user50000';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EXPLAIN
-> Index lookup on users using idx_username (...

Result Grid | Form Editor | Field Types | Query Stats

Result 5 | Read Only

SQLAdditions

Jump to

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Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	11:37:24	SELECT * FROM users WHERE username = 'user5000' LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
7	11:37:53	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.032 sec / 0.000 sec
8	11:38:04	EXPLAIN SELECT * FROM users WHERE username = 'user50000'	1 row(s) returned	0.015 sec / 0.000 sec
9	11:38:25	CREATE INDEX idx_username ON users (username)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.156 sec
10	11:38:42	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
11	11:39:02	EXPLAIN SELECT * FROM users WHERE username = 'user50000'	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1   SELECT * FROM users WHERE username LIKE 'user5%';
2 • EXPLAIN SELECT * FROM users WHERE username LIKE 'user5%';
3 • SELECT * FROM users WHERE id > 50000;
4 • EXPLAIN SELECT * FROM users WHERE id > 50000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EXPLAIN
-> Filter: (users.id > 50000) (cost=10002 rows...

Result Grid | Form Editor | Field Types | Query Stats

users 6 | Result 7 | users 8 | Result 9 | Read Only

SQLAdditions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	11:38:42	SELECT * FROM users WHERE username = 'user50000' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
11	11:39:02	EXPLAIN SELECT * FROM users WHERE username = 'user50000'	1 row(s) returned	0.000 sec / 0.000 sec
12	11:40:01	SELECT * FROM users WHERE username LIKE 'user5%' LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
13	11:40:01	EXPLAIN SELECT * FROM users WHERE username LIKE 'user5%'	1 row(s) returned	0.000 sec / 0.000 sec
14	11:40:01	SELECT * FROM users WHERE id > 50000 LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
15	11:40:01	EXPLAIN SELECT * FROM users WHERE id > 50000	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL97 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT
- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE
- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE
- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

Limit to 1000 rows

```
1 • DROP TABLE users;
2   DROP DATABASE index_performance_lab;
```

SQLAdditions

Jump to

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	11:40:01	SELECT * FROM users WHERE username LIKE 'user5%' LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
13	11:40:01	EXPLAIN SELECT * FROM users WHERE username LIKE 'user5%'	1 row(s) returned	0.000 sec / 0.000 sec
14	11:40:01	SELECT * FROM users WHERE id > 50000 LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
15	11:40:01	EXPLAIN SELECT * FROM users WHERE id > 50000	1 row(s) returned	0.000 sec / 0.000 sec
16	11:40:25	DROP TABLE users	0 row(s) affected	0.015 sec
17	11:40:25	DROP DATABASE index_perfomance_lab	0 row(s) affected	0.000 sec